RDS Table to HBase Table

Step 1 – use the following set of commands at root user to set up sqoop.

wget https://cdn.mysql.com//Downloads/Connector-J/mysql-connector-j-9.1.0.tar.gz

tar -xvf mysql-connector-j-9.1.0.tar.gz

sudo cp mysql-connector-j-9.1.0/mysql-connector-j-9.1.0.jar /usr/lib/sqoop/lib

Step 2 – Use the following command to ingest the data from MySQL RDS to Hbase table

sqoop import --connect jdbc:mysql://database-1.cdghlmkekpf2.us-east-1.rds.amazonaws.com/upgrad_assign --username admin --password admin123 --table yellow_taxi --hbase-table yellow_taxi_hbase --column-family cf1 --hbase-create-table --hbase-row-key VendorID,tpep_pickup_datetime,PULocationID,DOLocationID --hbase-bulkload --split-by payment_type

Explanation of the above command is as follows.

- `*--connect jdbc:mysql://database-1.cdghlmkekpf2.us-east-1.rds.amazonaws.com/upgrad_assign*` - Specifies the MySQL database connection URL pointing to an RDS instance in AWS. This tells Sqoop where to find the source database.
- `*--username admin --password admin123*` - These are the authentication credentials used to connect to the MySQL database. They provide the necessary access permissions to read from the source database.
- `*--table yellow_taxi*` - Identifies the source table name in MySQL that contains the data to be imported. This is where Sqoop will read the data from.
- `*--hbase-table yellow_taxi_hbase*` - Defines the destination table name that will be created in HBase. This is where the imported data will be stored.
- `*--column-family cf1*` - Specifies the column family name 'cf1' in HBase that will contain all the imported columns. In HBase, data is organized into column families for better organization and access.
- `*--hbase-create-table*` - This flag instructs Sqoop to automatically create the HBase table if it doesn't already exist. It saves the manual step of creating the table beforehand.
- `*--hbase-row-key VendorID,tpep_pickup_datetime,PULocationID,DOLocationID*` - Defines a composite row key using these four columns for uniquely identifying rows in HBase. This combination of fields will serve as the primary key.

- `*--hbase-bulkload*` - Enables bulk loading mode which writes directly to HBase files instead of using the HBase client API, resulting in faster data import performance.
- `*--split-by payment_type*` - Specifies the column used for partitioning the input data across multiple parallel mappers, helping to optimize the import process by enabling better parallelization.

```
root@ip-172-31-74-95:~
                Total vcore-milliseconds taken by all map tasks=1366585
                Total vcore-milliseconds taken by all reduce tasks=1116376
                Total megabyte-milliseconds taken by all map tasks=2099074560
                Total megabyte-milliseconds taken by all reduce tasks=3429507072
        Map-Reduce Framework
                Map input records=18880595
                Map output records=283208925
                Map output bytes=34611431988
                Map output materialized bytes=4629352281
                Input split bytes=591
                Combine input records=0
                Combine output records=0
                Reduce input groups=18843554
                Reduce shuffle bytes=4629352281
                Reduce input records=283208925
                Reduce output records=282653310
                Spilled Records=1018911100
                Shuffled Maps =5
                Failed Shuffles=0
                Merged Map outputs=5
                GC time elapsed (ms)=14283
                CPU time spent (ms)=2541720
                Physical memory (bytes) snapshot=4928999424
                Virtual memory (bytes) snapshot=21304528896
                Total committed heap usage (bytes)=4149739520
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=20869445040
25/01/12 21:20:47 INFO mapreduce.ImportJobBase: Transferred 19.4362 GB in 1,993.8042 seconds (9.9823 MB/sec)
25/01/12 21:20:47 INFO mapreduce.ImportJobBase: Retrieved 283208925 records.
25/01/12 21:20:47 WARN mapreduce.LoadIncrementalHFiles: managed connection cannot be used for bulkload. Creating unmanaged connection.
25/01/12 21:20:47 WARN mapreduce.LoadIncrementalHFiles: Skipping non-directory hdfs://ip-172-31-74-95.ec2.internal:8020/user/root/yellow_taxi/_SUCCESS
25/01/12 21:20:47 INFO impl.MetricsConfig: loaded properties from hadoop-metrics2-hbase.properties
25/01/12 21:20:47 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
25/01/12 21:20:47 INFO impl.MetricsSystemImpl: HBase metrics system started
25/01/12 21:20:47 WARN mapreduce.LoadIncrementalHFiles: Trying to bulk load hfile hdfs://ip-172-31-74-95.ec2.internal:8020/user/root/yellow_taxi/cf1/ef2c236
07d27480784f92f74d2b9bde8a with size: 11225589628 bytes can be problematic as it may lead to oversplitting.
25/01/12 21:20:47 INFO Configuration.deprecation: hbase.offheapcache.minblocksize is deprecated. Instead, use hbase.blockcache.minblocksize
[root@ip-172-31-74-95 ~]#
```

```
root@ip-172-31-74-95:~
hbase(main):002:0> scan 'yellow_taxi_hbase' {LIMIT+>2}
SyntaxError: (hbase):2: syntax error, unexpected tLCURLY

scan 'yellow_taxi_hbase' {LIMIT+>2}
                         ^

hbase(main):003:0> scan 'yellow_taxi_hbase', {LIMIT+>2}
SyntaxError: (hbase):3: syntax error, unexpected tGT

scan 'yellow_taxi_hbase', {LIMIT+>2}
                                  ^

hbase(main):004:0> scan 'yellow_taxi_hbase', {LIMIT=>2}
ROW                                    COLUMN+CELL
 1_2017-01-01 00:00:02.0_48_48         column=cf1:RatecodeID, timestamp=1736715786124, value=1
 1_2017-01-01 00:00:02.0_48_48         column=cf1:airport_fee, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:congestion_surcharge, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:extra, timestamp=1736715786124, value=0.5
 1_2017-01-01 00:00:02.0_48_48         column=cf1:fare_amount, timestamp=1736715786124, value=4.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:improvement_surcharge, timestamp=1736715786124, value=0.3
 1_2017-01-01 00:00:02.0_48_48         column=cf1:mta_tax, timestamp=1736715786124, value=0.5
 1_2017-01-01 00:00:02.0_48_48         column=cf1:passenger_count, timestamp=1736715786124, value=1
 1_2017-01-01 00:00:02.0_48_48         column=cf1:payment_type, timestamp=1736715786124, value=2
 1_2017-01-01 00:00:02.0_48_48         column=cf1:store_and_fwd_flag, timestamp=1736715786124, value=N
 1_2017-01-01 00:00:02.0_48_48         column=cf1:tip_amount, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:tolls_amount, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:total_amount, timestamp=1736715786124, value=5.3
 1_2017-01-01 00:00:02.0_48_48         column=cf1:tpep_dropoff_datetime, timestamp=1736715786124, value=2017-01-01 00:03:50.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:trip_distance, timestamp=1736715786124, value=0.5
 1_2017-01-01 00:00:03.0_162_161       column=cf1:RatecodeID, timestamp=1736715786124, value=1
 1_2017-01-01 00:00:03.0_162_161       column=cf1:airport_fee, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:03.0_162_161       column=cf1:congestion_surcharge, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:03.0_162_161       column=cf1:extra, timestamp=1736715786124, value=0.5
 1_2017-01-01 00:00:03.0_162_161       column=cf1:fare_amount, timestamp=1736715786124, value=6.0
 1_2017-01-01 00:00:03.0_162_161       column=cf1:improvement_surcharge, timestamp=1736715786124, value=0.3
 1_2017-01-01 00:00:03.0_162_161       column=cf1:mta_tax, timestamp=1736715786124, value=0.5
 1_2017-01-01 00:00:03.0_162_161       column=cf1:passenger_count, timestamp=1736715786124, value=1
 1_2017-01-01 00:00:03.0_162_161       column=cf1:payment_type, timestamp=1736715786124, value=1
 1_2017-01-01 00:00:03.0_162_161       column=cf1:store_and_fwd_flag, timestamp=1736715786124, value=N
 1_2017-01-01 00:00:03.0_162_161       column=cf1:tip_amount, timestamp=1736715786124, value=1.45
 1_2017-01-01 00:00:03.0_162_161       column=cf1:tolls_amount, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:03.0_162_161       column=cf1:total_amount, timestamp=1736715786124, value=8.75
 1_2017-01-01 00:00:03.0_162_161       column=cf1:tpep_dropoff_datetime, timestamp=1736715786124, value=2017-01-01 00:06:58.0
 1_2017-01-01 00:00:03.0_162_161       column=cf1:trip_distance, timestamp=1736715786124, value=0.8
2 row(s) in 0.1100 seconds

hbase(main):005:0>
```

```
root@ip-172-31-74-95:~

scan 'yellow_taxi_hbase', {LIMIT+>2}
                                  ^

hbase(main):004:0> scan 'yellow_taxi_hbase', {LIMIT=>2}
ROW                                    COLUMN+CELL
 1_2017-01-01 00:00:02.0_48_48         column=cf1:RatecodeID, timestamp=1736715786124, value=1
 1_2017-01-01 00:00:02.0_48_48         column=cf1:airport_fee, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:congestion_surcharge, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:extra, timestamp=1736715786124, value=0.5
 1_2017-01-01 00:00:02.0_48_48         column=cf1:fare_amount, timestamp=1736715786124, value=4.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:improvement_surcharge, timestamp=1736715786124, value=0.3
 1_2017-01-01 00:00:02.0_48_48         column=cf1:mta_tax, timestamp=1736715786124, value=0.5
 1_2017-01-01 00:00:02.0_48_48         column=cf1:passenger_count, timestamp=1736715786124, value=1
 1_2017-01-01 00:00:02.0_48_48         column=cf1:payment_type, timestamp=1736715786124, value=2
 1_2017-01-01 00:00:02.0_48_48         column=cf1:store_and_fwd_flag, timestamp=1736715786124, value=N
 1_2017-01-01 00:00:02.0_48_48         column=cf1:tip_amount, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:tolls_amount, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:total_amount, timestamp=1736715786124, value=5.3
 1_2017-01-01 00:00:02.0_48_48         column=cf1:tpep_dropoff_datetime, timestamp=1736715786124, value=2017-01-01 00:03:50.0
 1_2017-01-01 00:00:02.0_48_48         column=cf1:trip_distance, timestamp=1736715786124, value=0.5
 1_2017-01-01 00:00:03.0_162_161       column=cf1:RatecodeID, timestamp=1736715786124, value=1
 1_2017-01-01 00:00:03.0_162_161       column=cf1:airport_fee, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:03.0_162_161       column=cf1:congestion_surcharge, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:03.0_162_161       column=cf1:extra, timestamp=1736715786124, value=0.5
 1_2017-01-01 00:00:03.0_162_161       column=cf1:fare_amount, timestamp=1736715786124, value=6.0
 1_2017-01-01 00:00:03.0_162_161       column=cf1:improvement_surcharge, timestamp=1736715786124, value=0.3
 1_2017-01-01 00:00:03.0_162_161       column=cf1:mta_tax, timestamp=1736715786124, value=0.5
 1_2017-01-01 00:00:03.0_162_161       column=cf1:passenger_count, timestamp=1736715786124, value=1
 1_2017-01-01 00:00:03.0_162_161       column=cf1:payment_type, timestamp=1736715786124, value=1
 1_2017-01-01 00:00:03.0_162_161       column=cf1:store_and_fwd_flag, timestamp=1736715786124, value=N
 1_2017-01-01 00:00:03.0_162_161       column=cf1:tip_amount, timestamp=1736715786124, value=1.45
 1_2017-01-01 00:00:03.0_162_161       column=cf1:tolls_amount, timestamp=1736715786124, value=0.0
 1_2017-01-01 00:00:03.0_162_161       column=cf1:total_amount, timestamp=1736715786124, value=8.75
 1_2017-01-01 00:00:03.0_162_161       column=cf1:tpep_dropoff_datetime, timestamp=1736715786124, value=2017-01-01 00:06:58.0
 1_2017-01-01 00:00:03.0_162_161       column=cf1:trip_distance, timestamp=1736715786124, value=0.8
2 row(s) in 0.1100 seconds

hbase(main):005:0> describe 'yellow_taxi_hbase'
Table yellow_taxi_hbase is ENABLED
yellow_taxi_hbase
COLUMN FAMILIES DESCRIPTION
{NAME => 'cf1', BLOOMFILTER => 'ROW', VERSIONS => '1', IN_MEMORY => 'false', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', TTL => 'FOREVER',
 COMPRESSION => 'NONE', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE => '65536', REPLICATION_SCOPE => '0'}
1 row(s) in 0.0170 seconds

hbase(main):006:0>
```

Batch Ingestion

Step 1 – Ensure that happybase is installed.

sudo pip3 install happybase

Step 2 – Save the script as ***batch_ingest.py***

Step 3 – Run the script.

python3 batch_ingest.py

The script is written such that it will commit for every 10000 rows processed.

```
hadoop@ip-172-31-74-95:~

--2025-01-12 21:34:04--  https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-03.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 52.216.56.41, 52.216.204.147, 3.5.30.212, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|52.216.56.41|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 969809025 (925M) [text/csv]
Saving to: 'yellow_tripdata_2017-03.csv'

100%[===================================================================================================>] 969,809,025 28.1MB/s   in 28s

2025-01-12 21:34:32 (33.4 MB/s) - 'yellow_tripdata_2017-03.csv' saved [969809025/969809025]

[hadoop@ip-172-31-74-95 input_dataset]$ wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-04.csv
--2025-01-12 21:34:46--  https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-04.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 16.182.72.25, 52.217.122.161, 16.15.176.0, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|16.182.72.25|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 946349441 (903M) [text/csv]
Saving to: 'yellow_tripdata_2017-04.csv'

100%[===================================================================================================>] 946,349,441 31.9MB/s   in 25s

2025-01-12 21:35:12 (35.5 MB/s) - 'yellow_tripdata_2017-04.csv' saved [946349441/946349441]

[hadoop@ip-172-31-74-95 input_dataset]$ cd ..
[hadoop@ip-172-31-74-95 mr_assign]$ cd ..
[hadoop@ip-172-31-74-95 ~]$ python3 batch_ingest.py

Processing file: /home/hadoop/mr_assign/input_dataset/yellow_tripdata_2017-03.csv
Processed 10000 records
Processed 20000 records
Processed 30000 records
Processed 40000 records
Processed 50000 records
Processed 60000 records
Processed 70000 records
Processed 80000 records
Processed 90000 records
Processed 100000 records
Processed 110000 records
Processed 120000 records
Processed 130000 records
Processed 140000 records
Processed 150000 records
Processed 160000 records
Processed 170000 records
Processed 180000 records
Processed 190000 records
```

```
hadoop@ip-172-31-74-95:~

Processed 10040000 records
Processed 10050000 records
Processed 10060000 records
Processed 10070000 records
Processed 10080000 records
Processed 10090000 records
Processed 10100000 records
Processed 10110000 records
Processed 10120000 records
Processed 10130000 records
Processed 10140000 records
Processed 10150000 records
Processed 10160000 records
Processed 10170000 records
Processed 10180000 records
Processed 10190000 records
Processed 10200000 records
Processed 10210000 records
Processed 10220000 records
Processed 10230000 records
Processed 10240000 records
Processed 10250000 records
Processed 10260000 records
Processed 10270000 records
Processed 10280000 records
Processed 10290000 records
Finished processing 10295441 records from /home/hadoop/mr_assign/input_dataset/yellow_tripdata_2017-03.csv

Processing file: /home/hadoop/mr_assign/input_dataset/yellow_tripdata_2017-04.csv
Processed 10000 records
Processed 20000 records
Processed 30000 records
Processed 40000 records
Processed 50000 records
Processed 60000 records
Processed 70000 records
Processed 80000 records
Processed 90000 records
Processed 100000 records
Processed 110000 records
Processed 120000 records
Processed 130000 records
Processed 140000 records
Processed 150000 records
Processed 160000 records
Processed 170000 records
Processed 180000 records
```

```
hadoop@ip-172-31-74-95:~
Processed 9640000 records
Processed 9650000 records
Processed 9660000 records
Processed 9670000 records
Processed 9680000 records
Processed 9690000 records
Processed 9700000 records
Processed 9710000 records
Processed 9720000 records
Processed 9730000 records
Processed 9740000 records
Processed 9750000 records
Processed 9760000 records
Processed 9770000 records
Processed 9780000 records
Processed 9790000 records
Processed 9800000 records
Processed 9810000 records
Processed 9820000 records
Processed 9830000 records
Processed 9840000 records
Processed 9850000 records
Processed 9860000 records
Processed 9870000 records
Processed 9880000 records
Processed 9890000 records
Processed 9900000 records
Processed 9910000 records
Processed 9920000 records
Processed 9930000 records
Processed 9940000 records
Processed 9950000 records
Processed 9960000 records
Processed 9970000 records
Processed 9980000 records
Processed 9990000 records
Processed 10000000 records
Processed 10010000 records
Processed 10020000 records
Processed 10030000 records
Processed 10040000 records
Finished processing 10047135 records from /home/hadoop/mr_assign/input_dataset/yellow_tripdata_2017-04.csv

Sample data from HBase table:

Row key: 1_2017-01-01 00:00:02.0_48_48
cf1:RatecodeID: 1
```

```
hadoop@ip-172-31-74-95:~
Processed 10020000 records
Processed 10030000 records
Processed 10040000 records
Finished processing 10047135 records from /home/hadoop/mr_assign/input_dataset/yellow_tripdata_2017-04.csv

Sample data from HBase table:

Row key: 1_2017-01-01 00:00:02.0_48_48
cf1:RatecodeID: 1
cf1:airport_fee: 0.0
cf1:congestion_surcharge: 0.0
cf1:extra: 0.5
cf1:fare_amount: 4.0
cf1:improvement_surcharge: 0.3
cf1:mta_tax: 0.5
cf1:passenger_count: 1
cf1:payment_type: 2
cf1:store_and_fwd_flag: N
cf1:tip_amount: 0.0
cf1:tolls_amount: 0.0
cf1:total_amount: 5.3
cf1:tpep_dropoff_datetime: 2017-01-01 00:03:50.0
cf1:trip_distance: 0.5

Row key: 1_2017-01-01 00:00:03.0_162_161
cf1:RatecodeID: 1
cf1:airport_fee: 0.0
cf1:congestion_surcharge: 0.0
cf1:extra: 0.5
cf1:fare_amount: 6.0
cf1:improvement_surcharge: 0.3
cf1:mta_tax: 0.5
cf1:passenger_count: 1
cf1:payment_type: 1
cf1:store_and_fwd_flag: N
cf1:tip_amount: 1.45
cf1:tolls_amount: 0.0
cf1:total_amount: 8.75
cf1:tpep_dropoff_datetime: 2017-01-01 00:06:58.0
cf1:trip_distance: 0.8

Row key: 1_2017-01-01 00:00:05.0_48_50
cf1:RatecodeID: 1
cf1:airport_fee: 0.0
cf1:congestion_surcharge: 0.0
cf1:extra: 0.5
cf1:fare_amount: 7.0
```